

EXAMINATION INFORMATION PAGE

Written examination

Subject code: IIA4117		Subject name: Model Predictive Control	
Examination date: 28.11.2023	Examination time from/to: 09:00 – 12:00	Total hours: 3	
Responsible subject teacher: Roshan Sharma			
Campus: Porsgrunn		Faculty: Faculty of Technology	
No. of assignments: 3	No. of attachments: 0	No. of pages incl. front page and attachments: 3	
Permitted aids: None, only pen or pencil			
Information regarding attachments: No attachments			
Comments: Please use ball pen. This makes it easier for the examiner to read your answers. This exam counts 60% for the final grade.			

Select the type of examination paper

Spreadsheets

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Line sheets

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This exam counts 60% for the final grade.

Problem 1

Assume that you are going to design an MPC for controlling **a real industrial process** which has 4 states. Let us assume that you need to control all the 4 states of the system. Further, assume that all these 4 states of the system are measured (using sensors). You have two choices for controller design: (i) a state feedback MPC, and (ii) an output feedback MPC.

Tasks:

- i) [15%] For practical implementation, which one of the two controller design will you choose even though all 4 states are measurable? Why? Explain with proper justifications.
- ii) [5%] Explain briefly the difference between these two controller designs in general.

Problem 2

Consider the discrete time state space model of a process with disturbance acting on it.

$$x_{k+1} = Ax_k + Bu_k + B_d d_k \quad x_0 \text{ is known,} \quad (1)$$

$$y_k = Cx_k + C_d d_k \quad (2)$$

where the states are $x_k \in \mathbb{R}^{n_x}$, control inputs are $u_k \in \mathbb{R}^{n_u}$, process outputs are $y_k \in \mathbb{R}^{n_y}$, disturbances are $d_k \in \mathbb{R}^{n_d}$, $A \in \mathbb{R}^{n_x \times n_x}$, $B \in \mathbb{R}^{n_x \times n_u}$, $C \in \mathbb{R}^{n_y \times n_x}$, $B_d \in \mathbb{R}^{n_x \times n_d}$ and $C_d \in \mathbb{R}^{n_y \times n_d}$.

Tasks

- i) [15%] Considering that the disturbance acting on the system is constant or slowly varying, and both the system states and the disturbances are perfectly known or measured, how will you formulate an LQ optimal control problem for achieving integral action?
- ii) [5%] Now if we assume that the disturbances acting on the system are unknown and not directly measurable, can we still design an MPC with integral action? If yes, describe how this can be done.

Problem 3

Reduction of size of the optimal control problem using Lagrange multipliers is an effective way if the control problem only has equality constraints. Consider an unbounded LQ optimal control problem of the form,

$$\begin{aligned} \min_z \quad & \frac{1}{2} z^T H z + c^T z, \\ \text{s. t} \quad & A_\varepsilon z = b_\varepsilon, \end{aligned} \tag{3}$$

where z is a vector that contains the unknowns or decision variables.

Tasks:

- i) (15%) Explain in details how Lagrange multipliers can be used to reduce the size of the optimal control of equation (3).
- ii) (5%) What happens to the size of the unknowns to be optimized after the reduction? Comment about it.

Roshan Sharma
17th November 2023, Porsgrunn, Norway